

Multi-Agent Reinforcement Learning for Freshness-Aware Data Sensing Model in Vehicular Crowdsensing Systems

Penglin Dai^{1b}, Member, IEEE, Xin Wang^{1b}, Yue Xiang, Xiao Wu^{1b}, Member, IEEE, Junhua Wang^{1b}, and Kai Liu^{1b}, Senior Member, IEEE

Abstract—Vehicular Crowdsensing (VCS) is a promising paradigm for supporting urban sensing services, where Service Providers (SPs) engage Mobile Vehicles (MVs) to perform data sensing tasks with specific objectives. However, existing studies have predominantly focused on data sensing quality in terms of data collection completeness and geographic fairness, while largely neglecting the important aspect of data freshness. Moreover, effective mechanisms for optimizing data freshness through coordination of the behaviors of both SPs and MVs are still lacking. Accordingly, this paper proposes a Freshness-Aware Data Sensing (FDS) model by considering heterogeneous data freshness, varying sensing capabilities of MVs, and limited budgets of SPs. The FDS is formulated as a two-stage game model, where SPs and MVs iteratively determine their pricing and sensing strategies in a self-interested manner to maximize their individual gains. Further, we develop a multi-agent reinforcement learning-based approach to learn the pricing strategies based on historical observations, which allows SPs to make pricing decisions without global knowledge. Additionally, given the pricing strategies of SPs, the optimal solution for each MV is derived. Finally, we build the simulation model based on realistic vehicular traces, where the simulation results demonstrate the superiority of the proposed algorithm in various scenarios.

Index Terms—Freshness-aware data sensing, vehicular crowdsensing systems, multi-agent reinforcement learning, vehicular networks.

I. INTRODUCTION

WITH the advancements in vehicular sensing and communication technologies, Vehicular Crowdsensing (VCS)

has emerged as a promising paradigm for facilitating urban sensing services. The concept involves utilizing Mobile Vehicles (MVs) equipped with a variety of built-in sensors to carry out temporal data sensing tasks within urban areas, which are typically associated with a set of Points of Interest (PoIs). These MVs then wirelessly transmit the collected data to Service Providers (SPs) via nearby Base Stations (BSs). However, in dynamic environments, the content of the sensed data is subject to change over time, leading to a degradation in data freshness. This degradation can ultimately result in diminished user satisfaction. To ensure continuous participation of MVs in data sensing tasks, SPs must introduce incentive mechanisms that reward MVs based on their sensing behaviors. Conversely, the high mobility of MVs poses challenges as they may encounter unstable wireless connections and can only sense data from a PoI during a short dwelling time. Moreover, the limited wireless bandwidth and monetary budget further complicate the design of an optimal solution for both the data sensing strategies of MVs and the incentive mechanisms of SPs within VCS systems.

Recently, researchers have focused on various types of urban environmental data sensing for VCS applications, including traffic monitoring, parking recommendations, and High-Definition (HD) map collection. To address the diverse demands of VCS applications, studies have explored different VCS models, such as vehicle coverage [1] and cooperative versus non-cooperative game models [2]. Solutions for these models have been approached through optimization methods like submodular optimization [3] and evolutionary algorithms [4], typically focusing on objectives for either participants or recruiters. For joint optimization, two-stage strategies have been developed [5], which sequentially determine the sensing strategies of MVs and the recruiting strategies of SPs. Additionally, some studies have examined multifold performance aspects of VCS models, including energy consumption, communication overhead, geographic fairness, and data privacy.

Despite these efforts, the effect of data freshness in VCS systems remains underexplored. In fact, data freshness significantly impacts the performance of vehicular applications, such as Autonomous Driving (AD) and Vehicular Digital Twins (VDT). For example, [6] examined how data freshness affects autonomous driving systems, where a motion planner relies on temporal information provided by sensors. Their findings

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Penglin Dai, Xin Wang, Yue Xiang, and Xiao Wu are with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, Chengdu 611756, China, and with the Engineering Research Center of Sustainable Urban Intelligent Transportation, Ministry of Education, Beijing 100816, China, and also with the Tangshan Institute of Southwest Jiaotong University, Tangshan 063000, China (e-mail: penglindai@swjtu.edu.cn; wangxin000@my.swjtu.edu.cn; yxiang@my.swjtu.edu.cn; wuxiaohk@swjtu.edu.cn).

Junhua Wang is with the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China (e-mail: jhual207@nuaa.edu.cn).

Kai Liu is with the College of Computer Science, Chongqing University, Chongqing 400040, China (e-mail: liukai0807@cqu.edu.cn).

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indicate that as Age of Information (AoI) increases, vehicle safety is severely compromised, with collisions occurring when AoI surpasses certain critical thresholds [7]. Additionally, [8] proposed an enhanced AoI metric to assess data freshness in VDT and discovered that maintaining higher data freshness during the VDT synchronization process leads to improved Quality of Service (QoS). Existing solutions focus on traditional metrics such as data collection completeness and geographic fairness. They typically aim to optimize vehicle coverage over PoIs in a target area through vehicle selection [9] or task assignment [10] strategies. However, these approaches are not directly applicable for optimizing data freshness in VCS scenarios, where it is essential to examine individual vehicle sensing behaviors in relation to wireless resource allocation and competition. Regarding data freshness, existing studies use the AoI metric for evaluation in VCS applications [11]. However, several critical issues remain unresolved. First, these studies rely on the traditional AoI metric, which only measures the time elapsed since data generation and fails to capture the heterogeneity of different PoI types. Second, the relationship between the sensing behaviors of MVs and the data freshness gained by SPs is not yet well-defined. Third, while pricing mechanisms have been proposed for scenarios involving a single SP and multiple MVs, effective coordination between multiple SPs and MVs remains inadequately addressed.

To address these above issues, this work aims to optimize data freshness in VCS service scenarios, where multiple SPs keep the up-to-date status of data by recruiting MVs to perform environmental data sensing. Specifically, to address the first issue, we propose a new metric that combines the concept of AoI with varying decaying factors to capture the heterogeneity of PoIs, where the decaying factor is used to control the rate at which data freshness deteriorates over time and depends on PoI types. For the second issue, we develop the data sensing model based on wireless theory and derive a theoretical function to describe the relationship between the sensing power of MVs and the data freshness gained of SPs. To address the third issue, we introduce a pricing mechanism from SPs to incentivize MVs to participate in data sensing tasks. The coordination between multiple SPs and MVs is formulated as two-stage game model and the balanced solution is determined using a multi-agent reinforcement learning framework. Accordingly, the contributions of this work are outlined as follows.

- We investigate a novel Freshness-Aware VCS (FVCS) service scenario, where multiple SPs have to keep up-to-date temporal data of multiple PoIs by recruiting MVs for data sensing. Particularly, the data quality is comprehensively evaluated by synthesizing both the concept of AoI and varying decaying factors.
- We formulate the problem of Freshness-Aware Data Sensing (FDS) by developing a two-stage game model in the perspective of both SPs and MVs, respectively. In the first stage, SPs maximize the gains in terms of data freshness by making the pricing strategies with budget constraints. In the second stage, MVs maximize the gains in terms of the total revenues by adjusting their sensing behaviors within the constraints of transmission power. Both SPs and MVs

perform in a selfish way and aim at maximizing the gains of themselves.

- To resolve the FDS problem efficiently, we develop a multi-agent deep reinforcement learning algorithm for the pricing strategy, which enables each SP to make the optimal pricing decision independently based on its local observation without any prior knowledge of MVs and other SPs in dynamic VCS environment. Further, the optimal analytic solution of individual MV is derived based on the given pricing strategy of SPs.
- We build a simulation model by importing real world map, realistic vehicular traces and a python-programming based scheduling module. The comprehensive performance evaluations demonstrate the superiority of the proposed algorithm compared with two competitive solutions in a wide range of service scenarios.

The rest of this work is organized as follows. Section II gives the literature review. Sections III and IV formulate the system model and the FDS problem. In Section V, we propose the algorithm. Section VI gives the performance evaluation. Section VII makes a conclusion.

II. RELATED WORK

In the last decades, great efforts have been paid on improving the QoS performance of VCS systems in terms of different objectives. Some researchers focused on minimizing the energy consumption. [12] developed an energy-efficient framework, which makes task assignments and transfers sensing results in parallel with phone calls for minimizing both the number of anonymous participants and the waste of energy due to redundant task assignments. [13] formulated a novel multi-vehicle and multi-task offloading problem, and developed an online task scheduling algorithm based on Lyapunov optimization, which aims at minimizing the energy consumption of network-wide recruited MVs serving heterogeneous crowdsensing application. Some researchers focused on optimizing transmission performance during data collection in VCS systems. [14] presented an edge-assisted data uploading architecture, where MVs can be served by heterogeneous networks simultaneously. They developed an effective network selection and traffic allocation scheme based on particle swarm optimization, which aims at maximizing user satisfaction synthesizing transmission delay and bandwidth utilization. [15] presented a cooperative data forwarding mechanism to stimulate the selfish nodes to participate in multi-hop data forwarding using direct observations and mobile crowdsourcing information, which considerably improves network performance significantly in terms of data delivery ratio, transmission cost, and end-to-end delay. Data quality is one of the most critical QoS metrics on the performance evaluation of VCS systems. [16] proposed a quality-based truth estimation and surplus sharing method for quantifying the users' data qualities and long-term reputations and exploited an outlier detection technique to filter out anomalous data items, which achieves good performance in terms of both quality estimation and surplus sharing. [17] developed a mechanism to guarantee and enhance the quality of crowdsensing data in terms of accuracy

and privacy-preserving. The simulation results demonstrate the effectiveness of the proposed mechanism on data quality improvement and the privacy protection under extensive real-world traces.

Further, different types of models have been developed for formulating the VCS systems from different perspectives. Some researchers formulated the VCS models as participant selection problems from the perspective of SPs. [18] proposed a participant selection framework of SP, named CrowdRecruiter, which selects a small number of participants for minimizing incentive payments while satisfying probabilistic coverage constraint. [19] modeled the process of user recruitment as a stochastic submodular maximization problem, and designed a random adaptive greedy algorithm to guarantee a constant approximation ratio. [20] investigated an incentive mechanism and formulated a quality-aware pricing problem, which chooses an appropriate posted price to recruit a group of participants with reasonable sensing qualities and minimum expected payment. [21] investigated a multi-task allocation problem, which assigns real-time sensing task to the workers with limited available time for maximizing the utility of SP. However, these models assumed that MVs are always willing to participate in data sensing tasks, while neglecting the associated costs. Accordingly, some researchers have developed the VCS models from the view of both SPs and MVs. [22] formulated a vehicular crowdsensing game, where each participating MV chooses its sensing strategy based on the sensing cost, radio channel state, and the expected payment, and SP pays according to the accuracy of sensing report. The Nash equilibrium was derived and showed the tradeoff between the sensing accuracy and the overall payment. [5] proposed a two-stage game payoff decision-making scheme to solve the cooperation dilemma between SPs and MVs. They proposed a dynamic payment method to determine whether the platform adopts a cooperative strategy and developed a reward distribution method to motivate MVs to complete tasks with high quality and on time, which enhances MVs' QoS and decreases SPs' cost. [23] formulated a multi-stage Stackelberg game to model the competitive interactions between SPs as the leaders player and MVs as the followers. They derived the optimal sensing time allocation of MVs and iteratively adjusted the pricing strategies of SPs in a trial-and-error manner.

To solve these models, a lot of attentions have been paid on designing different types of optimization mechanisms. Some researchers formulated the VCS models as combinatorial optimization problems. For instance, [24] developed a Q-learning-based hyper-heuristic evolutionary algorithm by combining a memory-based initialization strategy and a strength-based neighborhood search, which effectively deals with a dynamic task allocation model in a self-learning way. Further, some researchers extended the VCS model as a non-linear or mixed non-linear optimization model. [14] decoupled a two-objective VCS problem into network selection and traffic allocation sub-problems, and derived an edge-based alternate search algorithm based on particle swarm optimization and convex optimization theories, which can jointly solve the two intertwined sub-problems. [25] studied the location-aware and location-diversity-based dynamic crowdsensing system, where

workers move over time and tasks arrive stochastically. They proposed a combinatorial algorithm for efficiently distributing tasks to workers and an online control policy based on Lyapunov optimization for workers' fair allocation. These methods are centralized-based scheduling and assumes that the global knowledge is known in advance, which is not suitable for non-cooperative environment. [26] proposed a Deep Reinforcement Learning (DRL)-based centralized control, distributed execution framework for multi-task-oriented VCS, called DRL-MTVCS. The proposed framework included an asynchronous architecture with spatiotemporal state information modeling, multi-task-oriented value estimate by adaptive normalization, and auxiliary vehicle action exploration by pixel control, which aims at minimizing simultaneously maximize the data collection amount, geographic fairness, and minimize the energy consumption of all vehicles for all tasks. [27] formulated the crowdsensing system as a multi-leader and multi-follower Stackelberg game, where Task Initiators (TIs) are the leaders and Mobile Users (MUs) are the followers. They proposed a DRL-based dynamic incentive mechanism, which enables TIs to learn the optimal pricing strategies directly from game experiences without knowing the private information of MUs. Previous studies [28] provide a comprehensive introduction and explanation of the application of hierarchical non-cooperative game theory in wireless communications and networking.

In contrast to previous research, our work explores the novel issue of FDS within VCS systems. This problem represents a unique combination of crowdsensing and network resource management, which differs from the traditional objectives of optimizing wireless communication performance. The freshness of data is evaluated by considering the sensing behaviors of MVs, AoI of the temporal data, and the heterogeneity among PoIs. Additionally, a two-stage game is formulated to model the interactions between SPs and MVs, aiming to maximize their individual objectives. To tackle this, this paper proposes a multi-agent reinforcement learning approach that learns the optimal solution in a centralized manner but implements scheduling at individual SP in a decentralized manner. This work is the first comprehensive study that addresses the FVCS model by characterizing the heterogeneity of data freshness among different PoIs, the capabilities of MVs for sensing data, and the coordination of pricing strategies among multiple SPs.

III. SYSTEM MODEL

Fig. 1 depicts a novel service architecture of the FVCS system. First, there exists multiple PoIs distributed among road networks, where each one is characterized by different data types and varying decaying factors. For instance, the chemical factory and the road segment are two types of PoIs with different data types. The former is related to the air quality index and the later is related to traffic flow. The traffic flow changes faster than air quality index. Therefore, we design a new metric to evaluate the freshness of heterogeneous temporal data. Second, there exists multiple SPs providing different kinds of services, such as traffic management, environmental monitoring and crowd density estimation, etc., where each SP requires to keep the

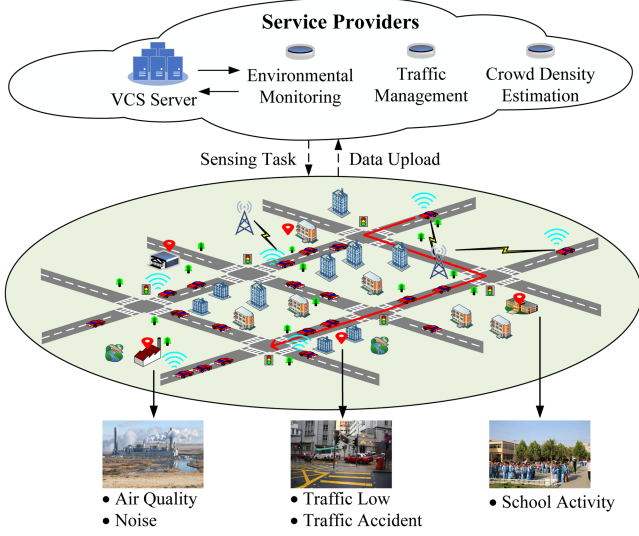


Fig. 1. The service architecture of the FVCS system.

up-to-date data from some set of PoIs by recruiting MVs for performing data sensing tasks. To motivate MVs, each SP makes its individual pricing strategy to reward the participated MVs based on both the quality and quantity of uploaded sensing data. Third, each MV is equipped with various types of sensors and is capable of sensing data when it enters the area of PoIs. Based on the price offered by SPs, each MV determines its data sensing behavior and subsequently uploads the sensed data by accessing a nearby BS. It should be noted that BSs are responsible of providing wireless network resources to facilitate the data upload from MVs. In contrast, SPs aggregate this data from various BSs to deliver VCS services. Additionally, SPs are responsible for paying the costs associated with incentivizing MVs to carry out data sensing tasks.

Particularly, the pricing strategies of SPs and the data sensing solutions of MVs are alternatively scheduled in a per time-slot way and the scheduling decisions of both SPs and MVs maintain unchanged within one time slot. First, each SP makes its own pricing strategy within budget constraint based on its own observation. Since both SPs and MVs are not willing to reveal their own decisions, SPs perform in a non-cooperative way. Second, based on the given pricing strategies of all SPs, each MV adjusts its wireless transmission power, which determines the data sampling rate. Accordingly, the data freshness of each PoI kept by SPs is determined by the joint uploading behaviors of all MVs within its area. Similarly, each MV performs in a selfish way and only concerns about maximizing its own gain.

IV. PROBLEM FORMULATION

In this section, we formulate the problem of Freshness-Aware Data Sensing (FDS) in the perspective of both SPs and MVs, which consists of two parts: Vehicular Crowdsensing (VCS) model and Service Provider (SP) model, respectively.

TABLE I
SUMMARY OF MAIN NOTATIONS

Notation	Description
$\mathbf{P}$	Set of PoIs
$\mathbf{V}$	Set of MVs
$\mathbf{B}$	Set of BSs
d_p	Temporal data of PoI p
s_p	Data size of d_p
$\mathbf{V}_p(t)$	Set of MVs in the area of PoI p at t
$\mathbf{P}_b$	Set of PoIs covered by BS b
$r_v(t)$	Data upload rate of MV v at t
$h_v(t)$	Transmission power of MV v at t
$V_b(t)$	Number of MVs accessing to BS b at t
$g_v(t)$	Wireless channel gain of MV v at t
h_v^{max}	Maximum transmission power of MV v
$\lambda_v(t)$	Data sampling rate of MV v at t
$c_s^p(t)$	Unit price SP s pays for PoI p at t
η_v	Unit sensing cost of MV v
$\mathcal{G}_v$	Gain of MV v
$\mathbf{P}_s$	Set of PoIs kept by SP s
C_s^{max}	Limited monetary budget of SP s
$\lambda_p(t)$	Data sampling rate of PoI p at t
$\mathcal{A}_p(t)$	AoI of data d_p kept by SPs at t
$g_p(t)$	Data freshness of PoI p at t
β_p	Decaying factor of data d_p
$g_p(t)$	Data freshness of PoI p during time slot $[t, t + \Delta t]$
ω_{sp}	Preference to PoI p showed by SP s
$\mathcal{G}_s(t)$	Gain of SP s during time slot $[t, t + \Delta t]$
$\mathcal{G}_s$	Gain of SP s
$\mathbf{C}_s$	Pricing matrix of SP s
$\mathbf{H}_v$	Transmission power vector of MV v
h_{vt}^*	Optimal transmission power of MV v at t

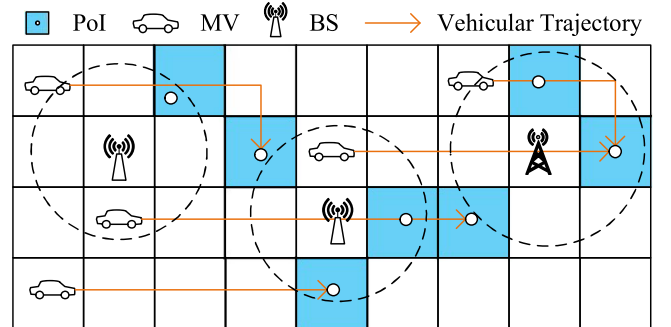


Fig. 2. The VCS model in the grid-based road map.

A. Vehicular Crowdsensing Model

We consider the scenario in a bounded 2D target region, where the whole region is divided into multiple identical grids. As shown in Fig. 2, there exists a set of PoIs denoted as $\mathbf{P} = \{1, 2, \dots, P\}$. Each PoI $p \in \mathbf{P}$ is assumed to occupy one grid and has data d_p with the size s_p . Further, a set of MVs denoted as $\mathbf{V} = \{1, 2, \dots, V\}$ is distributed among the road network and MVs can sample the up-to-date data when staying in any PoI $p \in \mathbf{P}$. Accordingly, vehicular trajectory is characterized by a series of notations $\mathbf{V}_p(t)$, $p \in \mathbf{P}$, $t = 1, 2, \dots, T$, where $\mathbf{V}_p(t)$ indicates the set of MVs dwelling in the area of PoI p at timestamp t . The primary notations are summarized in Table I.

The data sensed by MV v has to be uploaded to SPs via wireless communication. According to Fig. 2, there exists a set of BSs denoted as $\mathbf{B} = \{1, 2, \dots, B\}$, where each BS b covers

a set of PoIs denoted as $\mathbf{P}_b$ and provides data upload service for MVs in wireless communication range. Based on Shannon theory [29], given the wireless bandwidth W_b , the data upload rate of MV v at timestamp t , denoted as $r_v(t)$, is determined by the transmission power, denoted as $h_v(t)$, which is computed as follows.

$$r_v(t) = \frac{W_b}{V_b(t)} \log_2 \left(1 + \frac{g_v(t)h_v(t)}{\delta} \right), \quad (1)$$

where $V_b(t)$ is the number of MVs accessing to BS b at timestamp t , which indicates that the wireless bandwidth is allocated equally to each MV. Further, $g_v(t)$ denotes the wireless channel gain between MV v and BS b at timestamp t , which is computed by $g_v(t) = g_0|l_v - l_b|^{-\alpha}$, where g_0 represents the channel gain parameter, $|l_v - l_b|$ is the distance between MV v and BS b at timestamp t , and α denotes the path loss exponent. In addition, δ denotes the background noise power in communication channels. Let $h_v^{\max}$ denotes the maximum transmission power of MV v , then the transmission power of v at each timestamp t cannot exceed $h_v^{\max}$, i.e.,

$$h_v(t) \leq h_v^{\max}, t = 1, 2, \dots, T, \forall v \in \mathbf{V}. \quad (2)$$

In generally, data upload time is much larger than data sensing time. In this paper, data sampling time is neglected and the data upload follows zero-wait policy, which is a common setting in these literature [30], [31]. That is, the new data version will be sampled immediately when the previous version is completely uploaded. Therefore, for each MV v , the data sampling rate at timestamp t , denoted as $\lambda_v(t)$, is computed by the upload rate $r_v(t)$ divided by the data size s_p , which is formulated as follows.

$$\lambda_v(t) = \frac{r_v(t)}{s_p}, \forall v \in \mathbf{V}_p(t). \quad (3)$$

In addition, it is noted that the data upload behaviors of MVs are scheduled in each time slot $[t, t + \Delta t]$, where Δt denotes the constant duration of one time slot. The duration of Δt , determines the granularity level of the scheduling, which is tailored to the particular service scenario. High rates of change in environmental situation necessitate a shorter Δt to ensure timely updates of sensing strategy. Further, it is commonly assumed that events occur successively at each PoI and follows a Poisson process in these literature [32], [33]. As the data sampling from PoIs performs in a fixed time period, the data uploading behavior also follows a Poisson process according to the basic properties of the Poisson process.

In terms of individual MV, it is expected to receive payment from SPs as much as possible. However, it also has to consume the overhead caused by energy consumption during data sensing. Given the unit price that SP s pays for sensing data of PoI p at timestamp t , denoted as $c_s^p(t)$, and the unit sensing cost of MV v , denoted as η_v , the total gain of each MV v is defined as the total revenue minus the overhead caused on energy consumption, which is computed as follows.

$$\mathcal{G}_v = \sum_{t=1}^T \left(\sum_{\forall s \in \mathbf{S}} c_s^p(t) \lambda_v(t) - \eta_v h_v(t) \right) \Delta t. \quad (4)$$

It is observed that (4) is a non-linear function. To maximize the total revenue, each MV has to balance the trade-off between the revenue of data sensing and the overhead of energy consumption.

B. Service Provider Model

There exists a set of SPs $\mathbf{S} = \{1, 2, \dots, S\}$, where each SP s is expected to keep the update-to-date data from a subset of PoIs, denoted as $\mathbf{P}_s$. In particular, two SPs s and s' may require the data from some common PoIs, i.e., $\mathbf{P}_s \cap \mathbf{P}_{s'} \neq \emptyset$. It is assumed that both s and s' can receive the uploaded data from BS via wired or wireless connection. Further, each SP s has a limited monetary budget $C_s^{\max}$, which indicates that the total payoff for MVs during each time slot cannot exceed the maximum monetary budget $C_s^{\max}$, i.e.,

$$\sum_{\forall p \in \mathbf{P}_s} c_s^p(t) \lambda_p(t) \leq C_s^{\max}, t = 1, 2, \dots, T, \forall s \in \mathbf{S}, \quad (5)$$

where $\lambda_p(t)$ denotes the total data sampling rate of PoI p at timestamp t , which is computed by

$$\lambda_p(t) = \sum_{\forall v \in \mathbf{V}_p(t)} \lambda_v(t). \quad (6)$$

Setting a budget constraint for each time slot reduces the complexity of finding the optimal solution and aids in algorithm convergence, compared to an aggregated budget constraint. Additionally, assigning a budget $C_s^{\max}$ to each time slot allows for flexible adjustments to manage the overall budget by dividing the total budget constraint to the time periods.

Further, the AoI of data d_p from PoI p kept by SPs at timestamp t is defined as the current timestamp t minus the data generation time. Let $T_p(t)$ and $U_p(t)$ denote the latest upload time and the last data arrival timestamp of PoI p before timestamp t , then the data generation time is computed by $U_p(t) - T_p(t)$. Then, the AoI of data d_p kept by SPs at timestamp t , denoted as $\mathcal{A}_p(t)$, is computed as follows.

$$\mathcal{A}_p(t) = t + T_p(t) - U_p(t), \quad (7)$$

where $T_p(t)$ and $U_p(t)$ is determined by the sensing behaviors of all MVs $v \in \mathbf{V}_p(t)$ who participates in data sensing. As the data upload of individual MV follows a Poisson process, then according to the properties of the Poisson process, the arrival of $U_p(t)$ is also a Poisson process with parameter $\lambda_p(t)$. Accordingly, $T_p(t)$ follows the distribution of exponential distribution with parameter $\lambda_p(t)$.

Further, to assess data freshness for real-time applications, we design an information gain function that integrates the heterogeneity of PoI into the existing AoI-based utility function from [34], which is formulated as a negative exponential distribution with varying decaying factors. Then, the data freshness of PoI p benefit by SPs at an instantaneous t , denoted as $g_p(t)$, is computed as follows.

$$g_p(t) = e^{-\beta_p \mathcal{A}_p(t)}. \quad (8)$$

The decaying factor β_p for PoI p is determined based on its specific scenario. Generally, more urgent service scenarios require larger decaying factors, where the specific value can be fitted

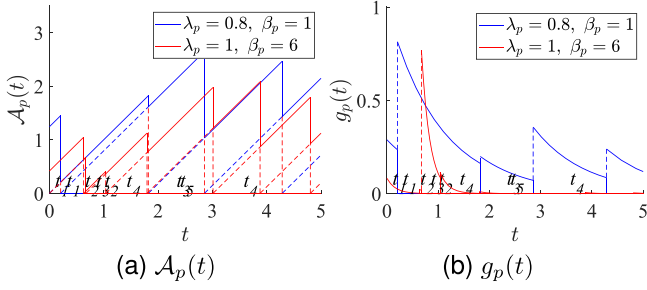


Fig. 3. The AoI $\mathcal{A}_p(t)$ and the data freshness $g_p(t)$ of two different Pols.

using historical data and statistics. Based on (8), the value of $g_p(t)$ naturally falls within the interval $[0,1]$. Then, β_p is the decaying factor of PoI p . Fig. 3 shows $\mathcal{A}_p(t)$ and $g_p(t)$ with different β_p and λ_p . It is noted that the decaying factor has great effect on $g_p(t)$. As shown in Fig. 3(a), it is validated by the fact that $\mathcal{A}_p(t)$ of p_2 with parameter $\lambda_p = 1$ (denoted by red curve) is lower than that of p_1 with parameter $\lambda_p = 0.8$ (denoted by blue curve). However, as shown in Fig. 3(b), $g_p(t)$ of p_2 is lower than p_1 since $\beta_p = 6$ of p_2 is higher than $\beta_p = 1$ of p_1 , which indicates that higher value of β_p will result in faster decaying rate of $g_p(t)$. The effect of β_p will be further discussed in Section VI.

It is noted that the value of $g_p(t)$ will decrease continuously with time elapsing, even within one time slot. Then, the expectation of data freshness of PoI p during time slot $[t, t + \Delta t]$, denoted as $\mathcal{G}_p(t)$, is computed as the average of the integral of $g_p(t)$ over $[t, t + \Delta t]$, which is formulated as follows.

$$\mathcal{G}_p(t) = \frac{1}{\Delta t} \int_t^{t+\Delta t} g_p(t) dt. \quad (9)$$

Further, each SP s shows different preferences to PoIs in the set $\mathbf{P}_s$. Let $\omega_{sp} \in [0,1]$ denotes the weight of SP s on PoI p , the gain of SP s during time slot $[t, t + \Delta t]$ is computed as the weighted summation of $\mathcal{G}_p(t)$ of $p \in \mathbf{P}_s$ and formulated as follows.

$$\mathcal{G}_s(t) = \sum_{p \in \mathbf{P}_s} \omega_{sp} \mathcal{G}_p(t), \quad (10)$$

where $\sum_{p \in \mathbf{P}_s} \omega_{sp} = 1$. ω_{sp} reflects SP's preferences for different PoIs, which can be determined by the popularity of the PoI. The value of ω_{sp} is higher for PoIs that attract interest from more service requesters, which can be estimated based on statistical data in practical applications.

Therefore, the total gain of each SP $s \in \mathbf{S}$ is defined as the summation of $\mathcal{G}_s(t)$ over time interval $[0, T]$, which is formulated as follows.

$$\mathcal{G}_s = \sum_{t=1}^T \mathcal{G}_s(t). \quad (11)$$

C. Two-Stage Game Model

In this section, we formulate the FDS problem as a two-stage game model in the perspective of both SPs and MVs, respectively, which is presented as follows.

In the first stage, each SP $s \in \mathbf{S}$ searches the pricing solution denoted by the pricing matrix $\mathbf{C}_s = [c_s^p(t)]_{T \times P}$, which aims at maximizing the objective function $\mathcal{G}_s$ within the budget constraint.

$$\begin{aligned} & \max_{\mathbf{C}_s} \mathcal{G}_s \\ \text{s.t. } & \sum_{p \in \mathbf{P}_s} c_s^p(t) \lambda_p(t) \leq C_s^{\max}, t = 1, 2, \dots, T, \forall s \in \mathbf{S}. \end{aligned} \quad (12)$$

In the second stage, each MV $v \in \mathbf{V}$ determines the transmission power denoted by the vector $\mathbf{H}_v = [h_v(t)]_{T \times 1}$, which aims at maximizing the objective function $\mathcal{G}_v$ within the constraint of maximum transmission power.

$$\begin{aligned} & \max_{\mathbf{H}_v} \mathcal{G}_v \\ \text{s.t. } & h_v(t) \leq h_v^{\max}, t = 1, 2, \dots, T, \forall v \in \mathbf{V}. \end{aligned} \quad (13)$$

Based on (12) and (13), both SPs and MVs perform in a selfish way and aim at maximizing their own objective. Therefore, they will not reveal their private information and the FDS problem is formulated as a non-cooperative game.

V. ALGORITHM DESIGN

In this section, we design the sensing solutions of MVs and the pricing strategies of SPs, respectively. First, the optimal analytic formulation of each MV is derived based on the given pricing of all SPs. Then, the pricing strategies of SPs are designed based on the framework of Multi-Agent Deep Deterministic Policy Gradient (MADDPG). The detail is presented as follows.

A. Optimal Solution of MV

According to the two-stage game model, each MV will decide its optimal transmission power based on the prices determined by SPs. Here, we formulate the optimal analytical solution for each MV by integrating its local knowledge with the strategic decisions of SPs. The theorem that follows establishes the optimal transmission power for each MV.

Theorem 1: For any MV $v \in \mathbf{V}_p(t)$, $\forall p \in \mathbf{P}$, given a feasible pricing solutions of all SPs on PoI p , denoted as $c_s^p(t)$, $\forall s \in \mathbf{S}$, the optimal transmission power of each MV v at timestamp t , denoted as h_{vt}^* , is derived as follows.

$$h_{vt}^* = \begin{cases} 0 & \text{if } h_{vt}^0 \leq 0, \\ h_v^{\max} & \text{if } h_{vt}^0 \geq h_v^{\max}, \\ h_{vt}^0 & \text{otherwise,} \end{cases} \quad (14)$$

where

$$h_{vt}^0 = \frac{K_v(t)}{\eta_v} - \frac{\delta}{g_v(t)}, K_v(t) = \frac{\sum_{s \in \mathbf{S}} c_s^p(t) W_b}{s_p V_b(t)}. \quad (15)$$

Proof: By using notation $K_v(t)$, the objective function $\mathcal{G}_v$ is simplified as

$$\mathcal{G}_v = \sum_{t=1}^T K_v(t) \log_2 \left(1 + \frac{g_v(t) h_v(t)}{\delta} \right) \Delta t - \sum_{t=1}^T \eta_v h_v(t) \Delta t. \quad (16)$$

Then, the partial derivative of $\mathcal{G}_v$ with respect to $h_v(t)$ is computed by

$$\frac{\partial \mathcal{G}_v}{\partial h_v(t)} = \left(\frac{K_v(t)g_v(t)}{g_v(t)h_v(t) + \delta} - \eta_v \right) \Delta t. \quad (17)$$

The solution of $\partial \mathcal{G}_v / \partial h_v(t) = 0$ is derived as

$$h_{vt}^0 = \frac{K_v(t)}{\eta_v} - \frac{\delta}{g_v(t)}. \quad (18)$$

Further, the second derivative of $\mathcal{G}_v$ with respect to $h_v(t)$ is computed by

$$\frac{\partial^2 \mathcal{G}_v}{\partial h_v(t)^2} = -\frac{K_v(t)g_v(t)^2 \Delta t}{(g_v(t)h_v(t) + \delta)^2} < 0, \quad (19)$$

which indicates that $\partial \mathcal{G}_v / \partial h_v(t)$ is a decreasing function with respect to $h_v(t) \in [0, h_v^{\max}]$. The discussion of the optimal transmission power, h_{vt}^* , will be divided into the following three cases.

- Case 1: $h_{vt}^0 < 0$ means that $\partial \mathcal{G}_v / \partial h_v(t) < 0$ with respect to $h_v(t) \in [0, h_v^{\max}]$, which indicates that $\mathcal{G}_v$ is a decreasing function with respect to $h_v(t) \in [0, h_v^{\max}]$. Accordingly, $h_{vt}^* = 0$ maximizes $\mathcal{G}_v$.
- Case 2: $h_{vt}^0 > h_v^{\max}$ means that $\partial \mathcal{G}_v / \partial h_v(t) > 0$ with respect to $h_v(t) \in [0, h_v^{\max}]$, which indicates that $\mathcal{G}_v$ is an increasing function with respect to $h_v(t) \in [0, h_v^{\max}]$. Accordingly, $h_{vt}^* = h_v^{\max}$ maximizes $\mathcal{G}_v$.
- Case 3: $h_{vt}^0 \in [0, h_v^{\max}]$ means that $\partial \mathcal{G}_v / \partial h_v(t) > 0$ with respect to $h_v(t) \in [0, h_{vt}^0]$ and $\partial \mathcal{G}_v / \partial h_v(t) < 0$ with respect to $h_v(t) \in [h_{vt}^0, h_v^{\max}]$, which indicates that $\mathcal{G}_v$ is an increasing function with respect to $h_v(t) \in [0, h_{vt}^0]$ and an increasing function with respect to $h_v(t) \in [h_{vt}^0, h_v^{\max}]$. Accordingly, $h_{vt}^* = h_{vt}^0$ maximizes $\mathcal{G}_v$.

Based on the above analysis, the optimal solution of each MV is derived as (14). $\square$

In summary, we utilize mathematical analysis techniques to calculate the first-order and second-order partial derivatives of the MV's gain, $\mathcal{G}_v$, with respect to the transmission power, $h_v(t)$. We then discuss the extremum points of the gain function within the power constraint on a case-by-case basis, which represents the optimal solution of transmission power and corresponding optimal gain of each MV. Consequently, we formulate a computational method for deriving the analytical solution of the optimal sensing strategy for each MV.

B. Pricing Strategy of SP

We design the MADDPG-based Pricing Strategy (MADDPG-PS) for SPs, which enables decentralized execution at each agent and centralized training by sharing extra information between different SPs. Before that, we first derive the expectation of data freshness of each PoI. Let Δt_n denotes the time interval between the $(n-1)_{th}$ and the n_{th} data uploads, then the data freshness of each PoI obtained by SPs during Δt_n is computed by the integral of $\mathcal{G}_p(t)$ over Δt_n , which is computed as follows.

$$\mathcal{G}_p(\Delta t_n) = \int_{t_{n-1}}^{t_n} e^{-\beta_p(t-t_{n-1})} dt = \frac{1}{\beta_p} (1 - e^{-\beta_p \Delta t_n}) \quad (20)$$

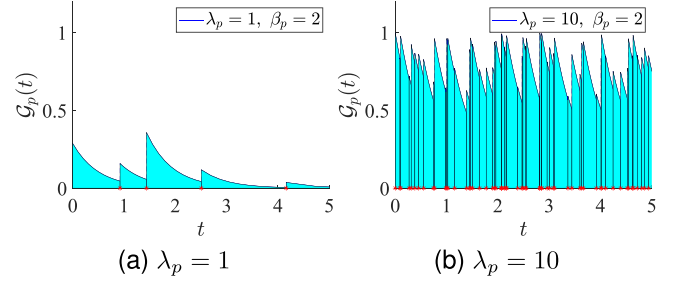


Fig. 4. The curve of $\mathcal{G}_p(t)$ with two different sampling rates.

It is assumed that the arrival of upload timestamps follow a Poisson process with parameter $\lambda_p(t)$, then the arrival time interval Δt_n follows the negative exponential distribution with parameter $\lambda_p(t)$. Accordingly, the expectation of $\mathcal{G}_p(\Delta t_n)$ is computed as follows.

$$\begin{aligned} \mathbb{E}(\mathcal{G}_p(\Delta t_n)) &= \frac{\lambda_p(t)}{\beta_p} \int_0^{+\infty} (1 - e^{-\beta_p \Delta t_n}) e^{-\lambda_p(t) \Delta t_n} d(\Delta t_n) \\ &= \frac{1}{\lambda_p(t) + \beta_p}. \end{aligned} \quad (21)$$

Based on the property of exponential distribution, the expectation of the arrival time interval Δt_n is $\mathbb{E}(\Delta t_n) = 1/\lambda_p(t)$. Then the expected data freshness of PoI p during time slot $[t, t + \Delta t]$ is computed by

$$\mathcal{G}_p(t) = \frac{\mathbb{E}(\sum_{n=1}^N \mathcal{G}_p(\Delta t_n))}{\mathbb{E}(\sum_{n=1}^N \Delta t_n)} = \frac{\lambda_p(t)}{\lambda_p(t) + \beta_p}. \quad (22)$$

Based on (22), the values of $\mathcal{G}_p(t)$ with two different sampling rates $\lambda_p(t)$ are shown in Fig. 4. It is observed that the value of $\mathcal{G}_p(t)$ is significantly lower when $\lambda_p(t) = 1$ compared to when $\lambda_p(t) = 10$, with the same $\beta_p = 2$.

Therefore, the total gain of each SP s is computed by the weighted summation of $\mathcal{G}_p(t)$, $\forall p \in \mathbf{P}_s$ over time interval $[0, T]$, which is formulated as

$$\mathcal{G}_s = \sum_{t=1}^T \sum_{p \in \mathbf{P}_s} \omega_{sp} \frac{\lambda_p(t)}{\lambda_p(t) + \beta_p}. \quad (23)$$

Then, the pricing procedure of each SP is modeled based on the framework of Markov Decision Process (MDP), where the basic elements are designed as follows.

- **Observation:** for each SP $s \in \mathbf{S}$, the observation at timestep t , denoted as o_s^t , is defined as the local information within its PoIs in the previous L time slots, which is formulated as

$$o_s^t = (\lambda_s^{t-L}, V_s^{t-L}, c_s^{t-L}, \dots, \lambda_s^{t-1}, V_s^{t-1}, c_s^{t-1}), \quad (24)$$

where λ_s^{t-l} , V_s^{t-l} and c_s^{t-l} denote the total data upload rate, the total MV number and the the total pricing paid by SP s at the $(t-l)_{th}$ time slot, respectively, which are computed

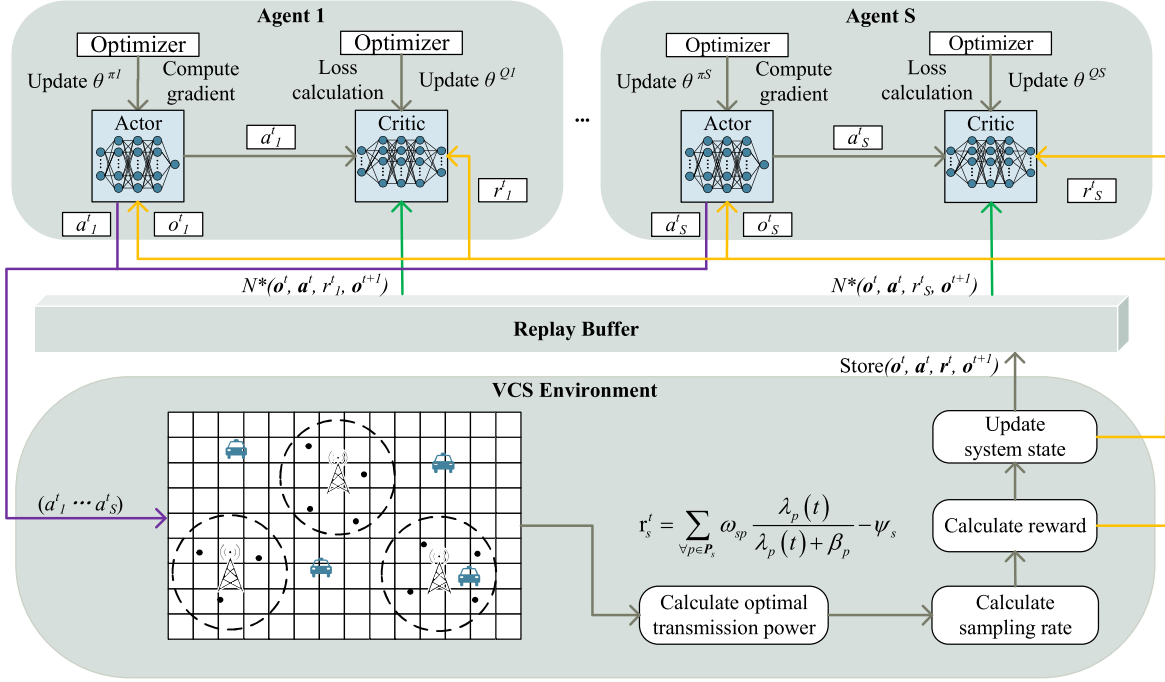


Fig. 5. The architecture of the proposed pricing strategy of SPs based on the framework of MADDPG.

by

$$\begin{aligned}\lambda_s^{t-l} &= \sum_{\forall p \in \mathbf{P}_s} \lambda_p(t-l), \\ V_s^{t-l} &= \sum_{\forall p \in \mathbf{P}_s} |\mathbf{V}_p(t-l)|, \\ c_s^{t-l} &= \sum_{\forall p \in \mathbf{P}_s} c_s^p(t-l).\end{aligned}\quad (25)$$

- **Action:** for each SP $s \in \mathbf{S}$, its action at timestep t is represented by a decision vector, which is denoted as

$$a_s^t = \{c_s^1(t), c_s^2(t), \dots, c_s^p(t), \dots, c_s^P(t)\}. \quad (26)$$

It is observed that in order to keep the same action dimension between different SPs, the dimension of each a_s^t is set to P . Particularly, for each SP s , the action related to unPoIs is set to zero, i.e., $c_s^p(t) = 0, \forall p \notin \mathbf{P}_s$.

- **Reward Function:** for each SP $s \in \mathbf{S}$, the reward function at timestep t , corresponding to the observation-action pair (o_s^t, a_s^t) , denoted as r_s^t , is defined as the weighted summation of the obtained gain related to data freshness from the set of PoIs minus a penalty function ψ_s , which is formulated as

$$r_s^t = \sum_{\forall p \in \mathbf{P}_s} \omega_{sp} \frac{\lambda_p(t)}{\lambda_p(t) + \beta_p} - \psi_s. \quad (27)$$

Let C_s^u denotes the budget utilization of SP s , which is defined as the ratio of the total payoff divided by the maximum budget and computed by

$$C_s^u = \frac{\sum_{\forall p \in \mathbf{P}_s} c_s^p(t) \lambda_p(t)}{C_s^{\max}}. \quad (28)$$

Then, the penalty function ψ_s is defined as follows.

$$\psi_s = \begin{cases} \kappa C_s^u & \text{if } C_s^u \geq 1, \\ 0 & \text{otherwise.} \end{cases} \quad (29)$$

where κ is a penalty parameter, which is adjustable to control the degree of the penalty imposed. The penalty function ψ_s , defined as a piecewise function, is used to penalize actions that violate budget constraints by subtracting a penalty term from the original reward. This design ensures that the reward function regulates action behaviors adhere to budget constraints through a learning-based approach.

On this basis, the architecture of MADDPG-PS is designed and shown in Fig. 5. Each SP $s \in \mathbf{S}$ is regarded as an individual agent and has its own actor network $\pi_s(o_s|\theta^{\pi_s})$ with weights θ^{π_s} , which is used for action generation based on its local observation. Further, each agent s has a critic network $Q_s(o, a|\theta^{Q_s})$ with θ^{Q_s} , which is used for policy evaluation based on a global view. According to Fig. 5, a replay buffer is used for storing the historical global experience by aggregating the local observations, actions and rewards of all SPs, which enables repeated training and expedites algorithm convergence. Further, to stabilize the training process, the target actor and critic networks are used for providing less volatile estimates of future actions and their values, where their parameters $\theta^{\pi'_s}$ and $\theta^{Q'_s}$ are duplicated from θ^{π_s} and θ^{Q_s} in a periodical way. Particularly, the scheduling procedure of the proposed pricing strategy comprises two stages: centralized training and decentralized execution.

First, in the training phase, for each agent s , the actor network $\pi_s(o_s|\theta^{\pi_s})$ and critic network $Q_s(o, a|\theta^{Q_s})$ with weights θ^{π_s} and θ^{Q_s} are randomly initialized. Then, the parameters of target actor and critic networks $\pi'_s(a_s|\theta^{\pi'_s})$ and $Q'_s(o, a|\theta^{Q'_s})$ are copied from original actor and critic networks, i.e., $\theta^{\pi'_s} \leftarrow \theta^{\pi_s}$

and $\theta^{Q'_s} \leftarrow \theta^{Q_s}$. Further, each agent s generates its action under its local observation o_s^t based on its actor network $\pi_s(o_s|\theta^{\pi_s})$ in a decentralized manner, i.e., $a_s^t = \pi_s(o_s^t|\theta^{\pi_s}) + \mathcal{N}^t$, where $\mathcal{N}^t$ is a noise. Each agent s executes its action and observes its own reward r_s^t based on (27) from the environment. All the local experiences $(o_s^t, a_s^t, r_s^t, o_s^{t+1})$, $\forall s \in \mathbf{S}$ are aggregated as $(\mathbf{o}^t, \mathbf{a}^t, \mathbf{r}^t, \mathbf{o}^{t+1})$ and stored in the global replay buffer $\mathcal{D}$. Further, each agent s samples mini-batch of N tuples $(\mathbf{o}^t, \mathbf{a}^t, \mathbf{r}^t, \mathbf{o}^{t+1})$ from $\mathcal{D}$ for model training, and computes target action value y_s^t as follows.

$$y_s^t = r_s^t + \gamma Q'_s(\mathbf{o}^{t+1}, \mathbf{a}^{t+1}|\theta^{Q'_s}), \quad (30)$$

where γ is the discount factor, reflecting the agent's emphasis on future rewards over immediate gains. The loss function of the critic network $Q_s(\mathbf{o}, \mathbf{a}|\theta^{Q_s})$ is computed by

$$\mathcal{L}(\theta^{Q_s}) = \frac{1}{N} \sum_t (Q_s(\mathbf{o}^t, \mathbf{a}^t|\theta^{Q_s}) - y_s^t)^2. \quad (31)$$

The critic network $Q_s(\mathbf{o}, \mathbf{a}|\theta^{Q_s})$ is updated by minimizing the loss function defined in (31), i.e.,

$$\theta^{Q_s} = \arg \min_{\theta^{Q_s}} \mathcal{L}(\theta^{Q_s}). \quad (32)$$

The objective function of the actor network is computed by

$$\mathcal{J}(\theta^{\pi_s}) = \frac{1}{N} \sum_t \pi_s(o_s^t|\theta^{\pi_s}) Q_s(\mathbf{o}^t, \mathbf{a}^t|_{a_s^t = \pi_s(o_s^t|\theta^{\pi_s})}). \quad (33)$$

Then, the actor network $\pi_s(o_s|\theta^{\pi_s})$ is updated by maximizing the objective function defined in (33), i.e.,

$$\theta^{\pi_s} = \arg \max_{\theta^{\pi_s}} \mathcal{J}(\theta^{\pi_s}). \quad (34)$$

Further, the target networks are updated based on the parameters from original actor and critic networks as follows.

$$\begin{aligned} \theta^{\pi'_s} &= \theta^{\pi_s} + (1 - \tau)\theta^{\pi'_s}, \\ \theta^{Q'_s} &= \theta^{Q_s} + (1 - \tau)\theta^{Q'_s}, \end{aligned} \quad (35)$$

where τ denotes the soft update parameter, which characterizes the magnitude of parameter adjustment. Second, after the policies have been trained, agents independently execute actions based on their own observations, no longer requiring the assistance of global information. They apply their learned policies to make real-time decisions, interacting directly with the environment. Overall, centralized training facilitates the policy learning of multiple SPs, introducing global information into each SP's critic network, enabling them to learn how to cooperate during training. Transitioning to distributed execution, SPs can autonomously respond to local changes without centralized control, which reflects the autonomy required in realistic scenarios and ensures real-time efficiency, crucial for swift decision-making in dynamic environments.

VI. PERFORMANCE EVALUATION

A. Default Setting

In this section, we implement the presented system model in Section III. Specifically, we use the realistic taxi traces from October 1, 2016, on the 5.6 km $\times$ 3.6 km area around the

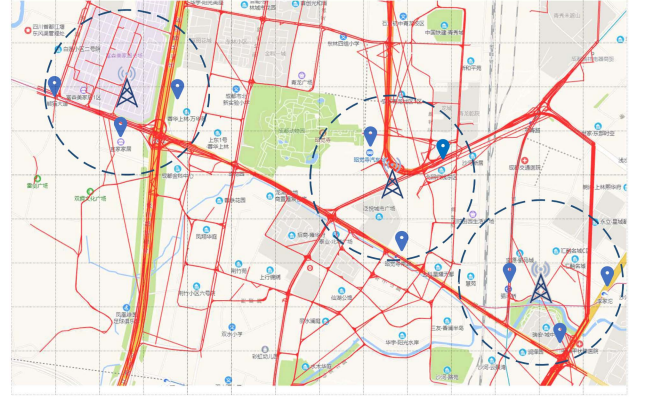


Fig. 6. The simulation scenario based on realistic vehicular traces.

TABLE II
DEFAULT PARAMETER SETTING

Parameter	Description	Value
$ \mathbf{S} $	Number of SPs	6
$ \mathbf{P} $	Number of PoIs	9
$ \mathbf{B} $	Number of BSs	3
$ \mathbf{V} $	Number of MVs	2958
h_v^{max}	Maximum transmission power of MV v	[11, 12]
η_v	Unit sensing cost of MV v	[0.03, 0.05]
s_p	Data size of PoI p	[15, 20]
β_p	Decaying factor of PoI p	[1, 3]
ω_{sp}	Weight of SP s on PoI p	$1 / \mathbf{P}_s $
C_s^{max}	Limited monetary budget of SP s	[14, 16]
W_b	Wireless bandwidth of BS b	[80, 100]
g_0	Channel gain parameter	10
α	Path loss exponent	3.2
δ	Background noise power	10^{-8}
Δt	Length of one time slot	30 sec

East Third Ring Road, Chengdu City, Sichuan Province, which is downloaded from Didi Chuxing GAIA Initiative accessed on [35]. As shown in Fig. 6, we divide the road map into 14×9 identical grids, where each grid is a square area with a side length of 350 meters. There exist 9 PoIs, indicated by distinctive blue beacon markers, and 3 BSs, symbolized by signal station icons, within the road network, with their respective positions being randomly assigned. Further, each BS is responsible for providing data upload service for MVs in its coverage, denoted by dotted blue line. Specifically, we selected 2958 MVs with 277,991 traces, which is highlighted in Fig. 6. For each MV v , the sensing cost η_v is randomly selected from the interval [0.03, 0.05], and the maximum transmission power h_v^{max} is selected from interval [11, 12]. For each PoI p , the data size is randomly selected from [15, 20] and the decaying factor is chosen from the interval [1, 3]. For each SP s , the budget constraint C_s^{max} is selected from [14, 16], and the weight on PoI p is $\omega_{sp} = 1/|\mathbf{P}_s|$, which expresses no preference for any specific PoI. For wireless communication, the bandwidth W_b , the channel gain g_0 , the path loss exponent α and the background noise power are set to 100, 10, 3.2 and 10^{-8} , respectively. Further, the length of each time slot, Δt , is set to 30 sec. These parameter settings are commonly used in literature [36], [37], [38]. Unless stated otherwise, the simulation is conducted under default setting listed in Table II.

For algorithm implementation, the network of each agent comprises one input layer, three hidden layers, and a single output layer. The hidden layers consist of 128, 64, and 32 neurons in sequence. Additionally, both the actor and critic networks employ the ReLU activation function for their hidden layers, while the actor network utilizes the Tanh function for the output layer to produce the policy outputs. Further, the specific parameters are set as follows. First, each episode is set to have a length of 30 timesteps, the observation of each agent is comprised of local information from the preceding 5 time slots, and the penalty parameter κ is set to 0.1. Then, the size of the replay buffer $\mathcal{D}$ is set to 10^5 . The discount factor γ , the soft update parameter τ , and the learning rate for the actor and critic networks, are set to 0.9, 0.01, and 1×10^{-3} , respectively. The mini-batch size N is set to 128, and the target networks synchronize with the original networks after every 2 updates. In addition, we employ an Ornstein-Uhlenbeck process ($\mu = 0, \theta = 0.15, \sigma = 0.2$) to generate correlated noise $\mathcal{N}$, which enhances training stability and policy update consistency, with the noise decaying as training progresses.

For performance comparison, as the proposed sensing solution is proved to be optimal, no alternative solutions are selected. With respect to the pricing strategy, we have adopted two competitive algorithms, which are presented as follows.

- Deep Deterministic Policy Gradient (DDPG) [39]: it is an actor-critic, model-free algorithm based on the deterministic policy gradient, which is implemented at each local SP. Accordingly, each SP makes its own pricing strategy independently based on its local observation.
- Deep Q-Network (DQN) [40]: it determines the optimal solution based on the value-action function approximated by deep neural networks. The DQN is implemented at each SP based on its local observation independently.

For performance evaluation, we collect the following statistics at each time slot $[t, t + \Delta t]$: the expectation of data freshness, $\mathcal{G}_p(t)$, the pricing decision $c_s^p(t)$, and the data sampling rate, $\lambda_p(t)$, of each PoI $p \in \mathbf{P}$, as well as the gain, $\mathcal{G}_v(t)$, and the actual transmission power $h_v(t)$, of each MV $v \in \mathbf{V}$. On this basis, we define the following metrics:

- Average Gain of SPs (AGSP): it is defined as the average gain of data freshness of all SPs across all PoIs over time interval $[0, T]$, which is computed by

$$\text{AGSP} = \frac{1}{T} \frac{1}{|\mathbf{S}|} \sum_{t=1}^T \sum_{\forall s \in \mathbf{S}} \frac{\sum_{\forall p \in \mathbf{P}_s} \mathcal{G}_p(t)}{|\mathbf{P}_s|}. \quad (36)$$

- Average Gain of MVs (AGV): it is defined as the average gain of all participating MVs over time interval $[0, T]$, which is computed by

$$\text{AGV} = \frac{1}{T} \frac{1}{|\mathbf{P}|} \sum_{t=1}^T \sum_{\forall p \in \mathbf{P}} \frac{1}{|\mathbf{V}_p(t)|} \sum_{\forall v \in \mathbf{V}_p(t)} \mathcal{G}_v(t). \quad (37)$$

- Average Budget Utilization (ABU): it is defined as the average ratio of the payment to the budget constraint of

all SPs over time interval $[0, T]$, which is computed by

$$\text{ABU} = \frac{1}{T} \frac{1}{|\mathbf{S}|} \sum_{t=1}^T \sum_{\forall s \in \mathbf{S}} \frac{\sum_{\forall p \in \mathbf{P}_s} c_s^p(t) \lambda_p(t)}{C_s^{\max}}. \quad (38)$$

- Average Power Utilization (APU): it is defined as the average ratio of transmission power consumed to the maximum transmission power available of all participating MVs over time interval $[0, T]$, which is computed by

$$\text{APU} = \frac{1}{T} \frac{1}{|\mathbf{P}|} \sum_{t=1}^T \sum_{\forall p \in \mathbf{P}} \frac{1}{|\mathbf{V}_p(t)|} \sum_{\forall v \in \mathbf{V}_p(t)} \frac{h_v(t)}{h_v^{\max}}. \quad (39)$$

B. Simulation Results

1) *Effect of Maximum Transmission Power*: In this section, we compare the performance of different algorithms under varying levels of maximum transmission power, denoted as $h_v^{\max}$. A higher value of $h_v^{\max}$ indicates that MVs possess stronger sensing capabilities, which may lead to a higher data sampling rate when coupled with an appropriate incentive mechanism. Fig. 7 illustrates the performance of three algorithms at different magnitudes of maximum transmission power, yielding the following observations. First, AGSP of all three algorithms exhibits a gradual increase as the maximum transmission power increases. This trend is evident in Fig. 7(a). Notably, MADDPG-PS achieves the highest AGSP across all cases among three algorithms. This can be attributed to the effective coordination of pricing strategies among multiple SPs by training local models based on shared global experiences. Fig. 7(c) provides evidence that MADDPG-PS not only secures a higher AGSP but also operates with a lower ABU than DDPG, which suggests that MADDPG-PS achieves greater benefits with a reduced budget. Second, among three algorithms, DDPG attains the highest AGV. This can be attributed to DDPG's focus on maximizing the local benefits of SPs by fully leveraging the sensing capabilities of MVs. Correspondingly, DDPG consistently achieves a higher APU than MADDPG-PS, as demonstrated in Fig. 7(d). However, it still lags behind MADDPG-PS in terms of AGSP due to its limited observation and inability to adequately balance budget distribution. Third, DQN performs poorly in AGSP and AGV due to its coarse reward evaluation and localized model training. MADDPG-PS, with centralized training, and DDPG, with finer granularity, both achieve better results.

The above observations can further be explained by the convergence performance of three algorithms. As shown in Fig. 8, MADDPG-PS achieves the best rewards in all cases. In addition, Fig. 8(a)~(c) demonstrate the rewards of three algorithms during different episodes, respectively. It is noted that MADDPG-PS also achieves the fast convergence and then maintains at a stable level. DDPG and DQN have slower convergence and still have some fluctuation. Fig. 9 shows the convergence performance of individual agent of three algorithms with maximum transmission power $h_v^{\max} \in [15, 16]$. As shown, MADDPG-PS elevates the smoothness of the updating process and the final performance of each agent, demonstrating superiority over DDPG and DQN in terms of stability and effectiveness, which further validates its

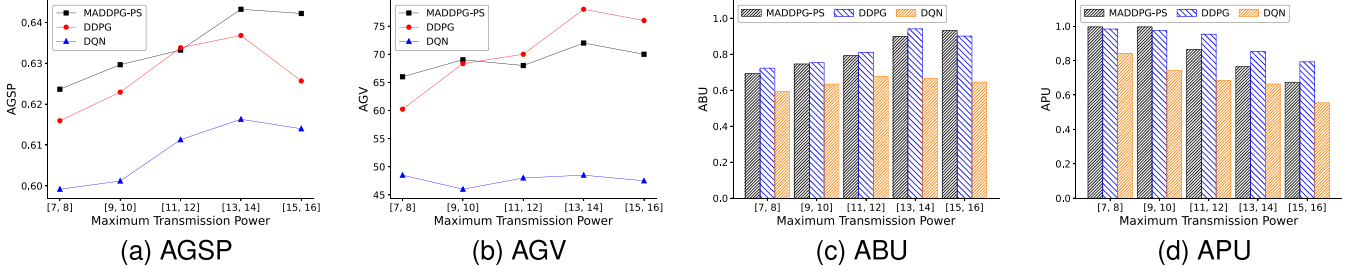


Fig. 7. The performance of three algorithms under varying maximum transmission powers.

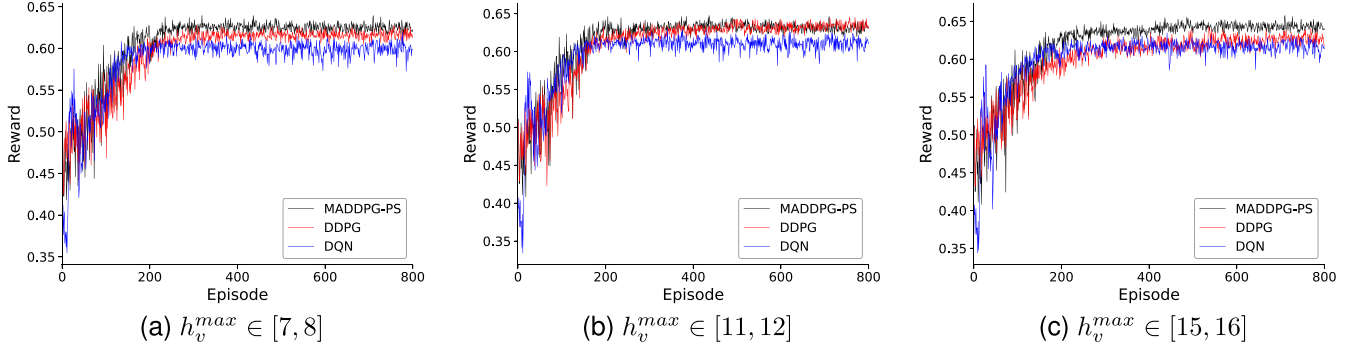


Fig. 8. The convergence performance of three algorithms under varying maximum transmission powers.

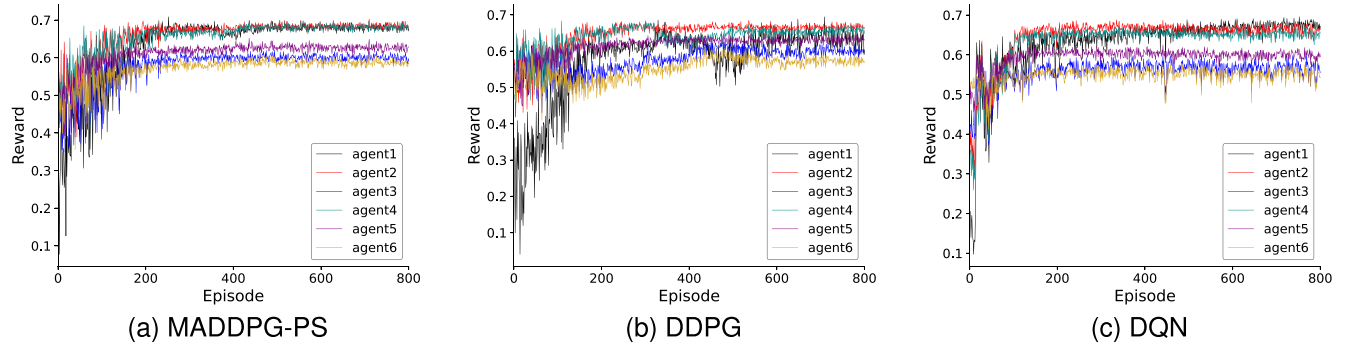


Fig. 9. The convergence performance of individual agent of three algorithms with $h_v^{\max} \in [15, 16]$.

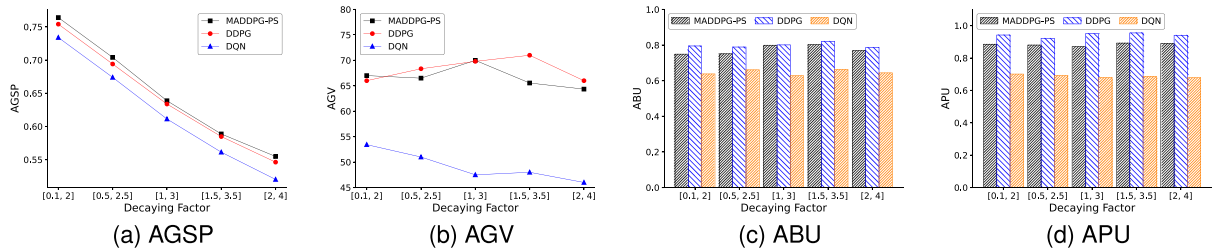


Fig. 10. The performance of three algorithms under varying decaying factors.

coordination capability. Based on the above analysis, this set of simulation results demonstrate the effectiveness of MADDPG-PS against different transmission powers.

2) *Effect of Decaying Factor*: In this section, we conduct a comparative analysis of algorithm performance under varying decaying factors. A higher decaying factor indicates a larger

degradation rate of data freshness. As illustrated in Fig. 10, it can be observed that AGSP of all three algorithms decreases significantly with an increase in the decaying factor. Further, we find that MADDPG-PS consistently achieves the highest AGSP among all three algorithms across all scenarios. This superior performance can be attributed to the reward performance of the

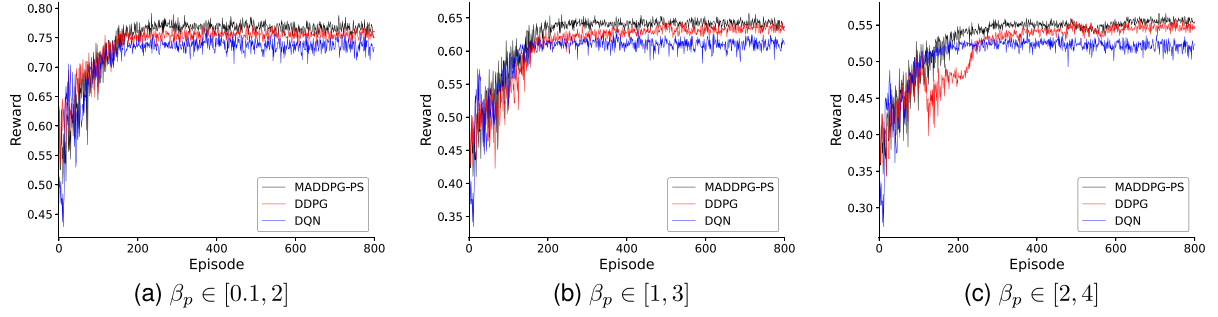


Fig. 11. The convergence performance of three algorithms under varying decaying factors.

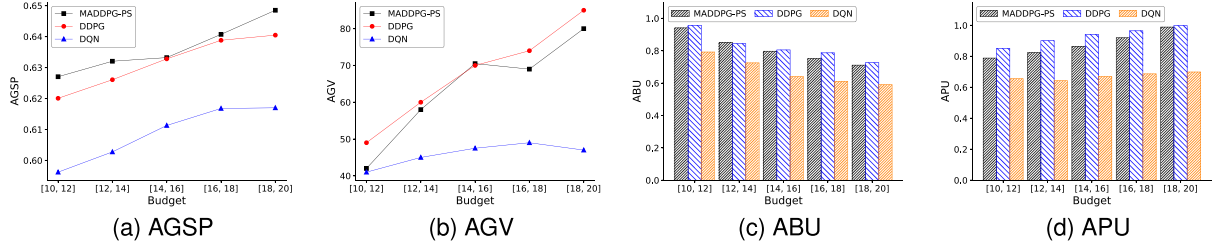


Fig. 12. The performance of three algorithms under varying budgets.

algorithm. As depicted in Fig. 11, MADDPG-PS demonstrates faster convergence compared to the other two algorithms and achieves the highest rewards. Notably, the reward of MADDPG-PS remains stable, while DDPG and DQN exhibit more fluctuations. In addition, MADDPG-PS and DDPG allocate a larger budget to motivate MVs to contribute more to sensing tasks. This is evident in Fig. 10(b)~(d), where AGV, ABU, and APU of MADDPG-PS and DDPG are significantly higher than those of DQN. Based on the observations, these results demonstrate the robustness of MADDPG-PS against varying decaying factors.

3) *Effect of Budget*: In this section, we compare the algorithm performance under varying budgets. An increased budget allows SPs to create more compelling incentives, thereby encouraging MVs to enhance their participation and contributions to data sensing activities. This trend is illustrated in Fig. 12(a), (b) and (d), showing that as the budget escalates, both AGSP, AGV and APU rise for all three algorithms. Further, according to Fig. 12(c), it is noted that ABU initially approaches 1, indicating that the budget is fully utilized. However, as the budget increases, its value gradually declines. This is because the budget was initially insufficient, but as it becomes more adequate, the relative utilization rate decreases. In addition, the DQN's performance is the worst in respective of AGSP and AGV, since it uses discrete action space instead, which cannot fully exploit the budget. In particular, MADDPG-PS consistently delivers the best performance across all scenarios, showcasing its superiority in various budgetary conditions.

4) *Effect of Channel Gain Estimation Error*: In this section, we conduct a performance evaluation under varying levels of channel estimation error, as detailed in Table III, where estimation errors of 1%, 5%, and 10% are introduced in the wireless channel gain. As shown, APU decreases progressively as the estimation error increases from 0 to 10%. This decline is primarily

TABLE III
PERFORMANCE COMPARISON WITH AND WITHOUT NOISE

	AGSP	AGV	ABU	APU
w/o noise	0.635	69.546	0.796	0.867
1% noise	0.632	69.542	0.796	0.861
5% noise	0.625	70.128	0.802	0.848
10% noise	0.620	70.502	0.809	0.823

due to the interference caused by the channel estimation error in the solution for $\mathcal{G}_v$ as indicated in (14). Consequently, more MVs' optimal transmission power falls below their maximum allowed power, leading to a reduction in AGSP as fewer vehicles participate, which results in a decrease in the freshness of the sensed data. Moreover, we observe a slight increase in ABU. This is likely due to SPs' enhanced efforts to incentivize greater participation, which modestly increases AGV. As a result, MVs benefit from higher rewards from SPs and experience reduced energy consumption cost.

VII. CONCLUSION

This paper primarily focuses on exploring the concept of freshness-aware data sensing in a heterogeneous VCS scenario, where the quality of data is characterized by temporal factors and data heterogeneity. To achieve this, an information gain function is introduced, which utilizes an exponential function based on the AoI and varying decaying factors. Subsequently, the problem of FDS is formulated as a two-stage game model, wherein both MV and SP make decisions regarding transmission power and pricing strategy, respectively, with the objective of maximizing their own gains. To address this problem, the paper proposes a pricing strategy framework based on the MADDPG-PS approach. This strategy enables each SP to execute actions based

on local observations in a decentralized manner, and the actor and critic networks are updated online using a global sharing replay buffer. This ensures fast convergence and adaptivity to dynamic VCS environments. Additionally, the optimal sensing solution for each MV is computed online based on the pricing strategy. Furthermore, a simulation model is built, incorporating real-world maps and realistic vehicular traces. The comprehensive performance evaluation demonstrates the fast convergence and superior performance of the proposed algorithm compared to other approaches like DDPG across various scenarios. It is important to note that the Poisson process does not capture all real-world scenarios, particularly those involving high mobility or bursty traffic. Therefore, in future work, we plan to investigate more dynamic and irregular traffic patterns, such as the Markov-Modulated Poisson Process (MMPP) and the Non-Homogeneous Poisson Process (NHPP). Specifically, we aim to explore data-driven approaches, including empirical distribution fitting and reinforcement learning, to model a wider variety of real-world traffic behaviors. Furthermore, we will investigate more detailed models of wireless interference and integrate fading effects to more accurately represent the time-varying nature of wireless channels, which is expected to improve the practical applicability of the proposed model.

Further, we discuss the potential for cooperation in our proposed FDS model. When the SPs and the MVs share a common goal, the cooperation can bring the mutual benefit for both of MVs and SPs. For example, in autonomous driving scenarios, both SPs and MVs need fresh data to ensure driving safety and efficient traffic management. In such cases, a cooperative game model can be effective. Conversely, when SPs and MVs have conflicting objectives, additional incentives, such as pricing mechanisms from SPs, may be necessary to encourage MVs' participation and achieve a balance between SPs and MVs.

REFERENCES

- [1] X. Chen et al., "PAS: Prediction-based actuation system for city-scale ridesharing vehicular mobile crowdsensing," *IEEE Internet Things J.*, vol. 7, no. 5, pp. 3719–3734, May 2020.
- [2] C.-L. Hu, K.-Y. Lin, and C. K. Chang, "Incentive mechanism for mobile crowdsensing with two-stage stackelberg game," *IEEE Trans. Serv. Comput.*, vol. 16, no. 3, pp. 1904–1918, May/Jun. 2023.
- [3] L. Liu, X. Wen, L. Wang, Z. Lu, W. Jing, and Y. Chen, "Incentive-aware recruitment of intelligent vehicles for edge-assisted mobile crowdsensing," *IEEE Trans. Veh. Technol.*, vol. 69, no. 10, pp. 12085–12097, Oct. 2020.
- [4] J. Ji, Y. Guo, D. Gong, and W. Tang, "MOEA/D-based participant selection method for crowdsensing with social awareness," *Appl. Soft Comput.*, vol. 87, 2020, Art. no. 105981.
- [5] H. Xia, R. Zhang, X. Cheng, T. Qiu, and D. O. Wu, "Two-stage game design of payoff decision-making scheme for crowdsourcing dilemmas," *IEEE/ACM Trans. Netw.*, vol. 28, no. 6, pp. 2741–2754, Dec. 2020.
- [6] C. Xu, Q. Xu, J. Wang, K. Wu, K. Lu, and C. Qiao, "AoI-centric task scheduling for autonomous driving systems," in *Proc. IEEE Conf. Comput. Commun.*, 2022, pp. 1019–1028.
- [7] S. Kaul, M. Gruteser, V. Rai, and J. Kenney, "Minimizing age of information in vehicular networks," in *Proc. 8th Annu. IEEE Commun. Soc. Conf. Sensor, Mesh Ad Hoc Commun. Netw.*, 2011, pp. 350–358.
- [8] L. Tang, Z. Cheng, J. Dai, H. Zhang, and Q. Chen, "Joint optimization of vehicular sensing and vehicle digital twins deployment for DT-assisted IoVs," *IEEE Trans. Veh. Technol.*, vol. 73, no. 8, pp. 11834–11847, Aug. 2024.
- [9] Z. Wang, Y. Cao, H. Zhou, L. Wu, W. Wang, and G. Min, "Fairness-aware two-stage hybrid sensing method in vehicular crowdsensing," *IEEE Trans. Mobile Comput.*, vol. 23, no. 12, pp. 11971–11988, Dec. 2024.
- [10] Q. Wu, Y. Li, G. Zhu, B. Mei, J. Xu, and M. Xu, "Prediction-aware adaptive task assignment for spatial crowdsourcing," *IEEE Trans. Mobile Comput.*, vol. 23, no. 12, pp. 13048–13061, Dec. 2024.
- [11] J. Li et al., "AoI-aware user service satisfaction enhancement in digital twin-empowered edge computing," *IEEE/ACM Trans. Netw.*, vol. 32, no. 2, pp. 1677–1690, Apr. 2024.
- [12] H. Xiong, D. Zhang, L. Wang, J. P. Gibson, and J. Zhu, "EEMC: Enabling energy-efficient mobile crowdsensing with anonymous participants," *ACM Trans. Intell. Syst. Technol.*, vol. 6, no. 3, pp. 1–26, 2015.
- [13] L. Pu, X. Chen, G. Mao, Q. Xie, and J. Xu, "Chimera: An energy-efficient and deadline-aware hybrid edge computing framework for vehicular crowdsensing applications," *IEEE Internet Things J.*, vol. 6, no. 1, pp. 84–99, Feb. 2019.
- [14] L. Liu, L. Wang, and X. Wen, "Joint network selection and traffic allocation in multi-access edge computing-based vehicular crowdsensing," in *Proc. IEEE Conf. Comput. Commun. Workshops*, 2020, pp. 1184–1189.
- [15] A. Rahim, K. Ma, W. Zhao, A. Tolba, Z. Al-Makhadmeh, and F. Xia, "Cooperative data forwarding based on crowdsourcing in vehicular social networks," *Pervasive Mobile Comput.*, vol. 51, pp. 43–55, 2018.
- [16] S. Yang, F. Wu, S. Tang, X. Gao, B. Yang, and G. Chen, "On designing data quality-aware truth estimation and surplus sharing method for mobile crowdsensing," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 4, pp. 832–847, Apr. 2017.
- [17] C. Zhao, S. Yang, and J. A. McCann, "On the data quality in privacy-preserving mobile crowdsensing systems with untruthful reporting," *IEEE Trans. Mobile Comput.*, vol. 20, no. 2, pp. 647–661, Feb. 2021.
- [18] D. Zhang, H. Xiong, L. Wang, and G. Chen, "Crowdrecruiter: Selecting participants for piggyback crowdsensing under probabilistic coverage constraint," in *Proc. ACM Int. Joint Conf. Pervasive Ubiquitous Comput.*, 2014, pp. 703–714.
- [19] S. Liu, Z. Zheng, F. Wu, S. Tang, and G. Chen, "Context-aware data quality estimation in mobile crowdsensing," in *Proc. IEEE Conf. Comput. Commun.*, 2017, pp. 1–9.
- [20] K. Han, H. Huang, and J. Luo, "Quality-aware pricing for mobile crowdsensing," *IEEE/ACM Trans. Netw.*, vol. 26, no. 4, pp. 1728–1741, Aug. 2018.
- [21] X. Li and X. Zhang, "Multi-task allocation under time constraints in mobile crowdsensing," *IEEE Trans. Mobile Comput.*, vol. 20, no. 4, pp. 1494–1510, Apr. 2019.
- [22] L. Xiao, T. Chen, C. Xie, H. Dai, and H. V. Poor, "Mobile crowdsensing games in vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 67, no. 2, pp. 1535–1545, Feb. 2018.
- [23] B. Gu, X. Yang, Z. Lin, W. Hu, M. Alazab, and R. Kharel, "Multiagent actor-critic network-based incentive mechanism for mobile crowdsensing in industrial systems," *IEEE Trans. Ind. Informat.*, vol. 17, no. 9, pp. 6182–6191, Sep. 2021.
- [24] J.-J. Ji, Y.-N. Guo, X.-Z. Gao, D.-W. Gong, and Y.-P. Wang, "Q-learning-based hyperheuristic evolutionary algorithm for dynamic task allocation of crowdsensing," *IEEE Trans. Cybern.*, vol. 53, no. 4, pp. 2211–2224, Apr. 2023.
- [25] X. Wang, R. Jia, X. Tian, and X. Gan, "Dynamic task assignment in crowdsensing with location awareness and location diversity," in *Proc. IEEE Conf. Comput. Commun.*, 2018, pp. 2420–2428.
- [26] C. H. Liu, Z. Dai, H. Yang, and J. Tang, "Multi-task-oriented vehicular crowdsensing: A deep learning approach," in *Proc. IEEE Conf. Comput. Commun.*, 2020, pp. 1123–1132.
- [27] Y. Zhan, C. H. Liu, Y. Zhao, J. Zhang, and J. Tang, "Free market of multi-leader multi-follower mobile crowdsensing: An incentive mechanism design by deep reinforcement learning," *IEEE Trans. Mobile Comput.*, vol. 19, no. 10, pp. 2316–2329, Oct. 2020.
- [28] Y. Zhang and M. Guizani, *Game Theory for Wireless Communications and Networking*. Boca Raton, FL, USA: CRC Press, 2011.
- [29] X. Chen, L. Jiao, W. Li, and X. Fu, "Efficient multi-user computation offloading for mobile-edge cloud computing," *IEEE/ACM Trans. Netw.*, vol. 24, no. 5, pp. 2795–2808, Oct. 2016.
- [30] Q. Kuang, J. Gong, X. Chen, and X. Ma, "Analysis on computation-intensive status update in mobile edge computing," *IEEE Trans. Veh. Technol.*, vol. 69, no. 4, pp. 4353–4366, Apr. 2020.
- [31] A. Alabbasi and V. Aggarwal, "Joint information freshness and completion time optimization for vehicular networks," *IEEE Trans. Serv. Comput.*, vol. 15, no. 2, pp. 1118–1129, Mar./Apr. 2022.
- [32] H. Gao et al., "Dynamic task pricing in mobile crowdsensing: An age-of-information-based queueing game scheme," *IEEE Internet Things J.*, vol. 9, no. 21, pp. 21278–21291, Nov. 2022.

- [33] J. Zhou, D. Li, and M. Liu, "BETA: From behavior sequentializing to task mapping in mobile crowdsensing," *IEEE Internet Things J.*, vol. 9, no. 19, pp. 18960–18972, Oct. 2022.
- [34] F. Li, Y. Sang, Z. Liu, B. Li, H. Wu, and B. Ji, "Waiting but not aging: Optimizing information freshness under the pull model," *IEEE/ACM Trans. Netw.*, vol. 29, no. 1, pp. 465–478, Feb. 2021.
- [35] "Data source: Didi chuxing gaia initiative," (n.d.), 2023. [Online]. Available: <https://gaia.didichuxing.com>
- [36] C. Liu, L. Wang, X. Wen, L. Liu, W. Zheng, and Z. Lu, "Efficient data collection scheme based on information entropy for vehicular crowdsensing," in *Proc. IEEE Int. Conf. Commun. Workshops*, 2022, pp. 1–6.
- [37] Y. Sun, S. Zhou, and Z. Niu, "Distributed task replication for vehicular edge computing: Performance analysis and learning-based algorithm," *IEEE Trans. Wireless Commun.*, vol. 20, no. 2, pp. 1138–1151, Feb. 2021.
- [38] Y. Zhang, X. Qin, and X. Song, "Mobility-aware cooperative task offloading and resource allocation in vehicular edge computing," in *Proc. IEEE Wireless Commun. Netw. Conf. Workshops*, 2020, pp. 1–6.
- [39] T. P. Lillicrap et al., "Continuous control with deep reinforcement learning," 2015, *arXiv:1509.02971*.
- [40] V. Mnih et al., "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.



Penglin Dai (Member, IEEE) received the BS degree in mathematics and applied mathematics and the PhD degree in computer science from Chongqing University, Chongqing, China, in 2012 and 2017, respectively. He is currently an associate professor with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, Chengdu, China. His research interests include Internet of vehicles, edge computing, and intelligent transportation systems.



Xin Wang received the BE degree in Internet of Things engineering from the Chongqing University of Posts and Telecommunications, Chongqing, China, in 2022. He is currently working toward the ME degree with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, Chengdu, China. His research interests include Internet of vehicles and collaborative perception.



Yue Xiang received the BE degree in spatial information and digital technology from the Chengdu University of Information Technology, China, in 2020. He is currently working toward the master's degree in electronic information with Southwest Jiaotong University. His research interests include mobile crowdsensing, internet of vehicles.



Xiao Wu (Member, IEEE) received the BEng and MS degrees in computer science from Yunnan University, Yunnan, China, in 1999 and 2002, respectively, and the PhD degree in computer science from the City University of Hong Kong, Hong Kong, in 2008. Currently, he is a professor and the assistant dean with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, Chengdu, China. He was with the Institute of Software, Chinese Academy of Sciences, Beijing, China, from 2001 to 2002. He was a research assistant and a senior research associate with the City University of Hong Kong, Hong Kong, from 2003 to 2004, and 2007 to 2009, respectively. He was with the School of Computer Science, Carnegie Mellon University, Pittsburgh, PA, USA, and with the School of Information and Computer Science, University of California, Irvine, CA, USA as a visiting scholar during 2006 to 2007 and 2015 to 2016, respectively. He has authored or co-authored more than 100 research papers in well-respected journals, such as *IEEE Transactions on Image Processing*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Medical Imaging* Peer-reviewed journal and prestigious proceedings like CVPR, ICCV and ACM MM. He received the Second Prize of Natural Science Award of the Ministry of Education, China, in 2016 and the Second Prize of Science and Technology Progress Award of Henan Province, China, in 2017. His research interests include artificial intelligence, computer vision, multimedia information retrieval, and image/video computing.



Junhua Wang received the BS and PhD degrees in computer science from Chongqing University, in 2014, and 2019, respectively. From 2017 to 2018, she was a visiting scholar with the University of Houston, USA. Since 2019, she has been working with the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics. Her research interests include mobile computing, vehicular adhoc networks, and wireless networks.



Kai Liu (Senior Member, IEEE) received the PhD degree in computer science from the City University of Hong Kong, in 2011. From December 2010 to May 2011, he was a visiting scholar with the Department of Computer Science, University of Virginia, USA. From 2011 to 2014, he was a postdoctoral fellow with Singapore Nanyang Technological University, City University of Hong Kong, and Hong Kong Baptist University. He is currently a professor with the College of Computer Science, Chongqing University, China. His research interests include Internet of vehicles, mobile computing and pervasive computing.